MIL-P-46295A (MU)
13 March 1964
SUPERSEDING
MIL-P-46295 (Ord)
21 October 1960

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M96
METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type of percussion primer designated as Primer, Percussion, M96.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition, General Specification for
MIL-I-45208 - Inspection Requirements, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105D - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1390

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DRAWINGS

ORDNANCE CORPS

7548516 - Primers, Percussion and Igniting, Details, Diaphragm
8837987 - Metal Parts Assembly
8838012 - Cup, Battery
8838015 - Plug, Firing

PUBLICATIONS

ORDNANCE CORPS

IEL-7157237 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Metal parts.-The metal parts shall comply with all requirements specified on Drawings (dwgs.) 8837987, 8838012 and 8838015, and with all requirements specified in applicable specifications.

3.3 Hydrostatic test.-The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.3.3.1 (see 6.5).

3.4 Strength of brazed joint.-The brazed joint securing the head and body shall withstand a pull of 5500 pounds without evidence of separation, when tested as specified in 4.3.3.2.

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3.5 Workmanship.-All parts shall be dry and free of cracks, chips, dust, grease, rust, burrs and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6.4. The reports shall contain, as a minimum, the following information:

a. Contractor and contract number.
b. Number and revision date of specification.
c. Characteristic code number.
d. Lot or serial number.
e. Number of defectives found.
f. Assignable cause or analysis of occurrence.
g. Corrective action taken or recommended. When the corrective action is beyond the control of the contractor

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or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 Initial production inspection

4.2.1 Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.6) for evaluation in accordance with the provisions of 4.2.2. The initial production sample shall consist of 25 complete sets of parts and 10 complete assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and requirements which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the inspections specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with any of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon failure of any component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

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4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of primers produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit as determined by the Government, shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL-STD-105D (or Standard MIL-STD-1235, when applicable).

4.3.2 Examination.-Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with Standard MIL-STD-105D except that continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.40 percent for each minor defect and an AQL of 0.25 percent for each major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.3.

4.3.2.1 Cup, battery (see dwg. 8838012).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Outside diameter, min.	Gage	01001
102.	Thickness through bottom, maximum (max.)	Gage	01002
103.	Length	Gage	01003
Minor:	AQL 2.50 percent		
201.	Diameter of hole	Gage	01004
202.	Thickness through bottom, min.	Gage	01005

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203.	Inside diameter	Gage	01006
204.	Outside diameter, max	Gage	01007
205.	Flat missing from surface at open end	Visual	01008
206.	Evidence of poor workmanship (see 3.5)	Visual	01009

4.3.2.2 Plug, firing (see dwg. 8838015).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Large diameter, min	Gage	02001
102.	Length and profile of point	Gage	02002
103.	Length of pilot, min	Gage	02003
104.	Total length	Gage	02004
Minor:	AQL 1.50 percent		
201.	Large diameter, max	Gage	02005
202.	Length of pilot, max	Gage	02006
203.	Diameter of pilot	Gage	02007
204.	Finish improper	Visual	02008
205.	Evidence of poor workmanship (see 3.5)	Visual	02009

4.3.2.3 Body (see dwg. 8838017 covering a detail of dwg. 8837987).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Flash hole missing	Gage	03001*
2.	Crack or split	Visual	03002*
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	03003
102.	Wall thickness, min	Gage	03004

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103.	Location of first two pairs of holes	Gage	03005
104.	Diameter of holes	Gage	03006
105.	Required plug gage fails to pass through body	Gage	03007
Minor:	AQL 1.50 percent		
201.	Wall thickness, max	Gage	03008
202.	Total length	Gage	03009
203.	Finish improper	Visual	03010
204.	Evidence of poor workmanship (see 3.5)	Visual	03011

4.3.2.4 Head (see dwg. 8837986 covering a detail of dwg. 8837987).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL .25 percent		
101.	Concentricity of flange with thread	Gage	04001
102.	Concentricity of body cavity with thread	Gage	04002
Minor:	AQL 2.50 percent		
201.	Depth of body cavity	Gage	04003
202.	Total length	Gage	04004
203.	Taper missing at base of body cavity	Visual	04005
204.	Chamber missing at face of body cavity	Visual	04006
205.	Evidence of poor workmanship (see 3.5)	Visual	04007
206.	Radii missing	Visual	04008
207.	Concentricity between percussion element cavity and thread	Gage	04009

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4.3.2.5 Metal parts assembly (see dwgs. 8837987 and 8837986).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Head assembled to wrong end of primer body	Visual	05001*
Major:	AQL 1.50 percent		
101.	Pitch diameter of thread	Gage	05002
102.	Major diameter of thread, min	Gage	05003
103.	Diameter of body, max	Gage	05004
104.	Diameter of undercut, max	Gage	05005
105.	Diameter of percussion element cavity	Gage	05006
106.	Depth of percussion element cavity from flange	Gage	05007
107.	Depth to percussion element cavity	Gage	05008
108.	Thickness of flange, max	Gage	05009
109.	Thread damaged	Visual	05010
110.	Crack or split in head	Visual	05011
111.	Flash hole in head obstructed	Visual	05012
112.	Braze incomplete or inadequate	Visual	05013
Minor:	AQL 6.50 percent		
201.	Major diameter of thread, max	Gage	05014
202.	Diameter of flange, max	Gage	05015
203.	Diameter of body, min	Gage	05016
204.	Concentricity between percussion element cavity and flash hole	Gage	05017
205.	Diameter of flash hole	Gage	05018
206.	Thickness of flange, min	Gage	05019

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207.	Length from flange to shoulder above thread	Gage	05020
208.	Width of thread undercut, max	Gage	05021
209.	Total length, max	Gage	05022
210.	Diameter of battery cup cavity	Gage	05023
211.	Depth of wrench slots	Gage	05024
212.	Distance between wrench slots	Gage	05025
213.	Diameter of wrench slots	Gage	05026
214.	Taper below percussion element cavity missing	Visual	05027
215.	Taper at mouth of percussion element cavity missing	Visual	05028
216.	Radii missing	Visual	05029
217.	Finish improper	Visual	05030
218.	Plating damaged exposing base metal	Visual	05031
219.	Evidence of poor workmanship (see 3.5)	Visual	05032

Note: When the alternative protective coating is used (Finish No. 1.9.2.3 or 1.1.2.3 of MIL-STD-171) dimensional inspection shall be performed on the assembly after plating and the AQL's of 4.3.2.5 shall apply to the combined defects.

4.3.2.6 Diaphragm (see dwg. 7548516).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.50 percent		
201.	Large outside diameter	Gage	06001
202.	Small outside diameter	Gage	06002
203.	Total length	Gage	06003

4.3.3 Testing.-The test equipment shall be in accordance with 4.3.4.

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4.3.3.1 Hydrostatic test (see 3.3) Critical defect - Code No. 07001*.-This test shall be conducted 100 percent. If any primer tubing fails to comply with the applicable requirements, it shall be classed defective and no primer body shall be fabricated therefrom. The test shall be performed as specified in 4.4.1.

4.3.3.2 Strength of brazed joint (see 3.4) Critical defect - Code No. 05033*.-A sample of 50 assemblies shall be selected from each lot for this test. If one or more assemblies fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.4 Inspection Equipment.-Index of Inspection Equipment Lists number IEL 7157237 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

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4.4 Test methods and procedures

4.4.1 Hydrostatic test.-Prior to machining, the primer tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

P = (2 St times .9) divided by D

Where:
P = test pressure in pounds per square inch (psi)
S = minimum specified yield strength, psi.
t = specified minimum wall thickness in inches.
D = specified maximum outside diameter in inches.

Observation shall be made for rupture or indication of flaws, as evidenced by leaks, and for permanent deformation beyond drawing requirements.

4.4.2 Strength of brazed joint.-The metal parts assembly shall be tested in a universal testing machine by applying the specified load in the axial direction. The rate of travel of the testing machine shall be 0.100 to 0.125 inch per minute. The metal parts assembly shall be observed for compliance with the applicable requirement.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-Packing and marking shall be in accordance with Specification MIL-P-10025.

5.2 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:
a. Title, number and date of this specification.

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6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 Inspection Equipment Lists.-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection Equipment List codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor's design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control Drawing.

6.3.3 Ordnance designs.-Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory.

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When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawings.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may

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be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608-12, and, in addition, contain the detail and information specified above.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Critical defect report

6.4.1 Submission of critical defect reports.-For U. S. Army Munitions Command procurements, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, Attn: SMUAP-RO; Picatinny Arsenal, Attn: SMUPA-ND.

6.4.2 Reporting to other services.-For other than U. S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the contracting officer.

6.5 Hydrostatic test.-If the primer tubing procured under this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

6.6 Submission of initial production samples.-Instruction as to the location for evaluation of initial production shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI.

Custodian:
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Preparing activity:
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